

PARVEEN KUMAR

Backend Engineer | Software Engineer | Full Stack Developer

📍 India | Parveen_24a12res426@iitp.ac.in | linkedin.com/in/parveen-kumar-664b8b24b | github.com/Parveen-collab | my-portfolio-black-chi-37.vercel.app

PROFESSIONAL SUMMARY

Backend Engineer with 1+ year of software development experience building scalable, production-ready systems using Java (Spring Boot), Node.js, Django, Go (Golang), and Python. Expertise spans RESTful API design, microservices architecture, JWT/OAuth2 authentication, MySQL/PostgreSQL/MongoDB databases, Docker, CI/CD pipelines, and AWS/Azure cloud deployment. Strong foundation in system design (HLD/LLD), distributed systems, and backend security. Complemented by full-stack proficiency in React.js and Next.js, and hands-on AI integration using OpenAI APIs and RAG architectures. Currently pursuing B.S. in Computer Science & Data Analytics at IIT Patna.

TECHNICAL SKILLS

Languages

Java • Python • Go (Golang) • JavaScript (ES6+) • TypeScript • SQL

APIs & Auth

REST API Design • JWT Authentication • OAuth2 • Microservices • API Gateway • WebSockets

Cloud & DevOps

Docker • Kubernetes • AWS • Azure • GitHub Actions • CI/CD • Linux • Nginx

AI Integration

OpenAI APIs • RAG Architecture • AI Agents • Prompt Engineering • LangChain

Backend Frameworks & Runtimes

Spring Boot • Spring Security • Hibernate/JPA • Node.js • Express.js • Django • Django REST Framework

Databases

MySQL • PostgreSQL • MongoDB • Redis • Query Optimization • Indexing • Schema Design

System Design

HLD / LLD • Scalability • Load Balancing • Caching Strategies • Distributed Systems • Design Patterns

Frontend & Tools

React.js • Next.js • Tailwind CSS • Git • Postman • IntelliJ IDEA • Figma

PROFESSIONAL EXPERIENCE

Frontend Developer Intern • Yuktisoft Infotech LLP

Jan 2025 – Present

- Developed and maintained scalable web application features using React.js, Next.js, TypeScript, and Material UI, serving 500+ daily active users.
- Integrated and consumed 15+ REST APIs (designed with Node.js/Express.js patterns) to power real-time data dashboards and reporting modules.
- Built reusable, accessible UI component library reducing development time for new features by ~30%.
- Implemented Django-backed data export endpoints (CSV/PDF) integrated into the frontend reporting pipeline.
- Participated in code reviews, debugging sessions, and performance optimization — improving page load speed by ~25% through lazy loading and caching strategies.
- Followed Git workflows, Agile/Scrum methodologies, and CI/CD best practices using GitHub Actions.

PROJECTS

Production-Scale Hiring Platform

Stack: Java • Spring Boot • Spring Security • Node.js • Hibernate/JPA • MySQL • PostgreSQL • Redis • Docker • AWS • React • Next.js • CI/CD

- Architected and built a production-scale hiring platform supporting employers, recruiters, and job seekers — designed to handle 10,000+ concurrent users via horizontal scaling.

- Implemented secure JWT + OAuth2 authentication and role-based access control using Spring Security, reducing unauthorized access incidents to zero in testing.
- Designed and exposed 40+ RESTful API endpoints for user management, job postings, applications, analytics, and notifications; documented with Swagger/OpenAPI.
- Built Node.js/Express.js microservice for real-time notification delivery (email + in-app) processing 1,000+ events/minute via message queues.
- Integrated Redis caching layer reducing average API response time by ~40% on high-traffic endpoints.
- Designed normalized MySQL + PostgreSQL schemas with optimized indexing; reduced complex query execution time by ~50%.
- Containerized all services with Docker and deployed on AWS using GitHub Actions CI/CD pipelines — achieving 99.9% uptime in staging environment.

Go-Based Microservices API Gateway

Stack: *Go (Golang) · gRPC · Docker · Redis · PostgreSQL · JWT*

- Designed and built a high-performance API gateway in Go (Golang) handling request routing, rate limiting, JWT validation, and load balancing across backend microservices.
- Implemented gRPC communication between Go services, achieving ~3x lower latency compared to equivalent REST-based inter-service calls.
- Applied Go concurrency patterns (goroutines, channels) to handle 5,000+ simultaneous connections with minimal memory overhead.
- Integrated Redis for distributed rate limiting and token caching; PostgreSQL for persistent audit logging.

Django REST Backend — Analytics Platform

Stack: *Python · Django · Django REST Framework · PostgreSQL · Celery · Redis · Docker*

- Built a Django REST Framework backend for a data analytics platform, exposing secure paginated APIs for dataset ingestion, processing, and visualization export.
- Implemented Celery + Redis async task queue to process large data jobs in the background, reducing API response time from ~8s to ~200ms.
- Designed PostgreSQL schema with materialized views and composite indexes; optimized slow queries achieving ~60% reduction in read latency.
- Secured APIs with Django's built-in auth + JWT token rotation; added rate limiting middleware and input validation to prevent injection attacks.
- Containerized with Docker Compose; wrote unit and integration tests achieving 85%+ code coverage.

NirmaanSetu – Global Construction Ecosystem Platform

Stack: *Next.js · TypeScript · Tailwind CSS · React · REST APIs*

- Contributed to a global-scale construction platform — built responsive frontend modules and designed scalable architecture for future backend integrations.
- Optimized web performance achieving Lighthouse scores of 90+ across all key pages.

CosmoVista – Space Exploration Platform

Stack: *Next.js · TypeScript · NASA APIs · REST*

- Built dynamic API-driven modules including Mars Rover Explorer, Asteroid Tracking, and NASA Media Integration using external REST APIs.

EDUCATION

B.S. — Computer Science & Data Analytics · Indian Institute of Technology (IIT) Patna

Expected July 2027

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Data Analytics, Numerical Methods.
- Applying backend engineering, system design, and data analytics skills through internship projects and independent full-stack development.

ACHIEVEMENTS & HIGHLIGHTS

- Built 3 production-ready backend systems using Java/Spring Boot, Node.js/Express.js, Django REST Framework, and Go — covering the full backend technology spectrum.

- Implemented AI integrations using OpenAI APIs, RAG architectures, and AI agents — delivering intelligent features in full-stack applications.
- Completed 1+ year of software development internship at Yuktisoft Infotech LLP with measurable impact on performance and development efficiency.
- Solved 100+ Data Structures & Algorithms problems across LeetCode and similar platforms for interview preparation.
- Maintained active GitHub portfolio with deployed applications, engineering projects, and open-source contributions.

ADDITIONAL INFORMATION

Languages: English (Professional) • Hindi (Native) • German (A1) **Core Competencies:** Backend Engineering • Microservices • REST API Design • System Design • Cloud Deployment • AI Integration • Full Stack Development